

Quant Developer

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- Systematic Portfolio Construction
 - Mean-Variance Optimization (MVO)
 - Disciplined Convex Programming
 - Signal Integration & Risk Overlays

Portfolio Variance: The Two-Asset Case

$$\sigma_p^2 = \mathbf{w}^T \Sigma \mathbf{w}$$

$$\sigma_p^2 = \begin{bmatrix} w_A & w_B \end{bmatrix} \begin{bmatrix} \sigma_A^2 & \rho \sigma_A \sigma_B \\ \rho \sigma_B \sigma_A & \sigma_B^2 \end{bmatrix} \begin{bmatrix} w_A \\ w_B \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} \text{Var}_A & \text{Cov}_{AB} \\ \text{Cov}_{AB} & \text{Var}_B \end{bmatrix}$$

- $\mathbf{w}$: Portfolio weights $\longrightarrow \mathbf{1}^T \mathbf{w} = 1$
- Σ : **Covariance matrix** (positive semi-definite)
- $\rho \in [-1, 1]$: Correlation $\longrightarrow$ reflect **actual risk** in portfolio
- $\sigma_i^2 > 0$: Single variance ($\sigma_i^2 = 0 \iff$ const. return, e.g. fixed rates)

Portfolio Optimization is Convex

Conditions for a Global Minimum:

- **First Order:** $\nabla f(\mathbf{w}) = 0$ (Stationary point)
 - **Second Order:** $\nabla^2 f(\mathbf{w}) = \mathbf{H} \succeq 0$ (Positive semi-definite Hessian)
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Proof: Positive Semi-Definiteness (PSD) and Eigenvalues

Let $\mathbf{v}$ be an eigenvector of Σ with eigenvalue λ :

$$\Sigma \mathbf{v} = \lambda \mathbf{v}$$

$$\mathbf{v}^T \Sigma \mathbf{v} = \mathbf{v}^T (\lambda \mathbf{v}) = \lambda (\mathbf{v}^T \mathbf{v}) = \lambda \|\mathbf{v}\|^2$$

Since $\|\mathbf{v}\|^2 \geq 0$, the quadratic form $\mathbf{v}^T \Sigma \mathbf{v} \geq 0$ if and only if $\lambda \geq 0$.

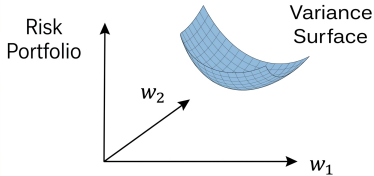
Summary for Portfolio Context:

- $\Sigma \succeq 0$ (PSD) $\iff$ All eigenvalues $\lambda_i \geq 0$
 - Variance $f(\mathbf{w}) = \mathbf{w}^T \Sigma \mathbf{w}$ is **convex** $\iff \Sigma$ is PSD
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Note: This property ensures that any local minimum is also the global minimum.

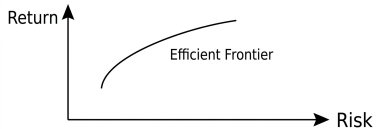
Visualizing Convex Portfolio Optimization

Variance Surface



- Quadratic form $\mathbf{w}^T \Sigma \mathbf{w}$
- Convex shape (bowl)
- Global minimum exists

Efficient Frontier (EF)



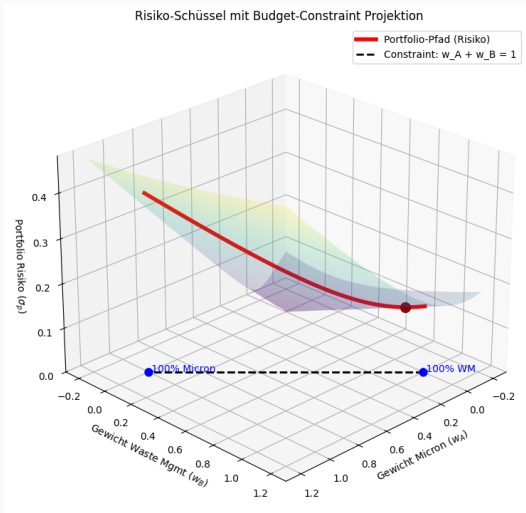
- Risk vs. Return trade-off
- Set of optimal portfolios
- Solved via convex optimization

EF: Corresponds to a cross-section of the variance surface

EF: Sequence of points providing maximum return for each risk level

Stefan: “Schnitt durch die Risiko-Schüssel”

Visualization of the Variance Surface

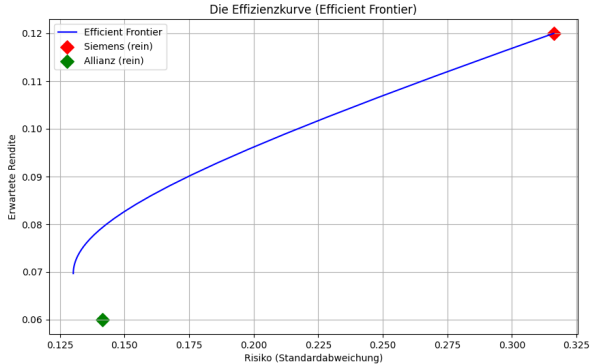


- $1 = w_{MU} + w_{WM}$
- Risk increases with higher **Micron** exposure
- Low risk due to significant **Waste Management** allocation

The convex shape guarantees a unique Minimum Variance Portfolio.

Note: Variance surface calculated via Python (CVXPY/NumPy); returns based on simulated price data.

Efficient Frontier



- **Left:** Minimum Variance Portfolio (MVP)
- **Right:** Maximum Return Portfolio
- **Below:** Inefficient portfolios (suboptimal)
- **Above:** Mathematically impossible region

Minimizing Portfolio Variance

$$\min_w \frac{1}{2} \sigma^2 = \min_w \frac{1}{2} \mathbf{w}^T \Sigma \mathbf{w} \quad (\text{Objective Function})$$

For two assets A and B , the variance is:

$$\sigma^2 = \sigma_A^2 w_A^2 + \sigma_B^2 w_B^2 + 2w_A w_B \rho \sigma_A \sigma_B$$

Definitions:

- $\rho = \frac{\text{Cov}(A, B)}{\sigma_A \sigma_B}$ (Normalization)
- $\text{Cov}(A, B) = \frac{1}{n-1} \sum_{i=1}^n (R_{A_i} - \bar{R}_A)(R_{B_i} - \bar{R}_B)$

Optimization (Lagrange):

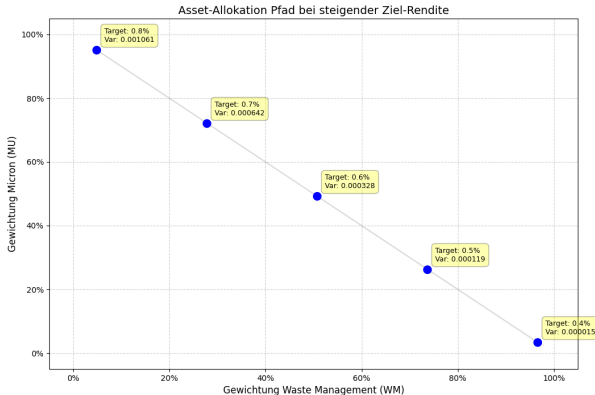
- $\frac{d\sigma^2}{dw_A} \stackrel{!}{=} 0$
- Subject to: $1 = w_A + w_B$

$$w_A = \frac{\sigma_B^2 - \text{Cov}(A, B)}{\sigma_A^2 + \sigma_B^2 - 2\text{Cov}(A, B)}$$

Interpretation:

1. If σ_B^2 is large $\rightarrow$ Numerator increases $\rightarrow$ Flight to asset A
2. If $\text{Cov}(A, B)$ becomes negative $\rightarrow$ Denominator increases $\rightarrow$ Diversification effect

Customer Case: Increasing Target Return



- **Shift in weights** due to higher target return
- Consequently portfolio **variance increases** by moving along EF

The optimizer requires a lower bound for returns: $\mathbf{w}^T \boldsymbol{\mu} \geq \mu_{\text{Target}}$

Relative Strength Index (RSI) - Momentum Indicator

```
python RSI_PayPal.py
```

*****100%*****

*****100%*****

1 of 1 completed

1 of 1 completed

Date	Kurs_USD	FX_Rate	Kurs_EUR	RSI	Signal
2026-01-14	57.49	1.16	49.38	43.11	HOLD
2026-01-15	56.57	1.16	48.57	35.31	HOLD
2026-01-16	56.72	1.16	48.86	37.79	HOLD
2026-01-20	54.91	1.16	47.18	32.22	HOLD
2026-01-21	55.72	1.17	47.51	35.90	HOLD
2026-01-22	56.98	1.17	48.81	46.25	HOLD
2026-01-23	56.45	1.18	48.02	43.70	HOLD
2026-01-26	56.43	1.19	47.59	34.86	HOLD
2026-01-27	55.34	1.19	46.60	28.96	BUY
2026-01-28	53.92	1.20	44.86	27.00	BUY
2026-01-29	52.95	1.20	44.21	25.82	BUY
2026-01-30	52.53	1.20	43.90	26.12	BUY
2026-02-02	52.17	1.18	44.03	27.37	BUY
2026-02-03	41.57	1.18	35.24	16.03	BUY
2026-02-04	40.91	1.18	34.62	10.85	BUY
2026-02-05	39.78	1.18	33.71	10.79	BUY
2026-02-06	40.30	1.18	34.22	11.83	BUY
2026-02-09	41.03	1.18	34.69	15.27	BUY
2026-02-10	41.36	1.19	34.75	13.95	BUY
2026-02-11	40.34	1.19	33.93	6.80	BUY
2026-02-12	38.96	1.19	32.81	6.67	BUY
2026-02-13	40.17	1.19	33.84	12.16	BUY
2026-02-17	40.69	1.18	34.34	15.28	BUY
2026-02-18	41.32	1.19	34.87	19.63	BUY
2026-02-19	41.60	1.18	35.29	22.52	BUY
2026-02-20	41.52	1.18	35.28	22.94	BUY
2026-02-23	43.92	1.18	37.11	30.36	HOLD
2026-02-24	46.88	1.18	39.75	69.67	HOLD
2026-02-25	47.18	1.18	40.06	74.38	SELL

$$RSI = 100 - \frac{100}{1 + RS}$$

$$RS = \frac{\frac{1}{n} \sum \text{Gains}}{\frac{1}{n} \sum |\text{Losses}|}$$

- Momentum indicators aim to predict **trend reversals**.
- A falling price often indicates deteriorating **expected returns** or shifting **correlations** within the portfolio.

⇒ Identifying oversold (< 30) and overbought (> 70) **not sufficient** for rebalancing

⇒ RSI ignores the portfolio's overall convexity

Convex vs. Concave Optimization - Two Equivalent Ways

Risk Minimization

Focus on the **Convex Bowl**:

$$\min \frac{1}{2} \mathbf{w}^T \Sigma \mathbf{w}$$

cp.**Minimize**(risk)

Utility Maximization

Focus on the **Concave Hill**:

$$\max \left(-\frac{1}{2} \mathbf{w}^T \Sigma \mathbf{w} \right)$$

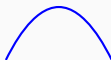
cp.**Maximize**(-risk)

DCP Rules (Disciplined Convex Programming):

- ⇒ **CVXPY** only allows cp.Maximize() for **concave** functions
- ⇒ Since risk is convex, negative risk (later utility) is concave
- ⇒ Both approaches yield the **exact same** optimal weights w



Convex (Min)



Concave (Max)

Objective: Utility Maximization with linear Risk and Tax

$$\max \left(\mu^T \mathbf{w} - \frac{\lambda}{2} \mathbf{w}^T \Sigma \mathbf{w} - \text{tax}^T \mathbf{w} \right)$$

```
w = cp.Variable(n)

p_return    = expected_returns @ w
p_risk      = cp.quad_form(w, cov_matrix)
tax_penalty = tax_impact @ w

utility = p_return - (lambda_risk / 2) * p_risk - tax_penalty

objective = cp.Maximize(utility)

constraints = [cp.sum(w) == 1, w >= 0, w <= 0.1 ]

prob = cp.Problem(objective, constraints)
prob.solve()
```

Summary & Outlook - Beyond Markowitz

- **Backward-looking:** Classic MPT relies solely on **historical data**
- **Instability:** Small return changes can lead to massive weight shifts
- **Missing Fundamentals:** Model ignores company's health

Thinking Out of the Box: Fund. Integration of **Equity Ratio**

Idea 1: Fundamental Covariance Σ_{Equity} (Risk Penalty):

$$U = \mu^T \mathbf{w} - \underbrace{\frac{\lambda}{2} \mathbf{w}^T \Sigma_{Return} \mathbf{w}}_{\text{Market Risk}} - \underbrace{\frac{\delta}{2} \mathbf{w}^T \Sigma_{Equity} \mathbf{w}}_{\text{Fundamental Risk}} - \text{Tax}^T \mathbf{w}$$

Idea 2: Equity Ratio as Linear Benefit:

$$U = \mu^T \mathbf{w} - \frac{\lambda}{2} \mathbf{w}^T \Sigma_{Return} \mathbf{w} + \gamma \text{Equity}^T \mathbf{w} - \text{Tax}^T \mathbf{w}$$

Idea 3: Equity Ratio as Constraint (e.g., $\text{Equity}^T \mathbf{w} \geq \text{threshold}$)

- + **Stability:** Fundamental ratios are far **less volatile**
- + **Resilience:** Provides a buffer against **shifting environments**